

Assignment Part I

This is the first part of the assignment for the course “Large Scale Optimization using Decomposition” (42136), Spring 2025. This project must be passed in order to be able to attend the written examination.

Some useful information

- This assignment can be done in groups of ideally **two** but at most **three** students
- The deadline for the assignment is **18/3**, 23:59
- Assignments (and any code) must be submitted via DTU Learn
- A forum is available on DTU Learn where you can ask questions. Only questions asked through the forum will be answered.
- All required data is available in the “**assignment-1-data.jl**” file on DTU Learn

The problem

Consider a company that must decide how much capacity to reserve in advance from a set of potential suppliers, I , in order to serve a set of clients, J , whose future demands are uncertain. Demand uncertainty is represented by a finite set of scenarios, S , each with a known probability of occurring, p_s , and a demand realization for every client, d_{js} .

For each supplier $i \in I$, the company pays a fixed reservation cost, f_i , if that supplier is used, as well as a per-unit cost of procured capacity c_i . Once a demand scenario occurs, the company may use the reserved capacity to serve client demands at a scenario-dependent operating cost t_{ijs} . If a client's demand cannot be fully satisfied in a given scenario, the unmet portion is penalized at a given (potentially customer-specific) cost, u_{js} . Each supplier has an upper limit, b_i , on how much capacity can be reserved from the supplier, and the company has an internal requirement that no more than n different suppliers can be used. The company must determine a capacity reservation plan that minimizes the expected total cost, including capacity procurement, operating costs, and penalties for unmet demand.

No specific units are provided for the quantity-related parameters in this assignment. Assume that all quantity parameters, such as capacities and scenario dependent client demands, are expressed in the same (unspecified) unit of measure.

All parameter values are made available in the “**assignment-1-data.jl**” file. Complete the tasks on the following pages.

Prepare a report of no more than **10 pages** that provides complete and well structured solutions to all tasks listed on the following page. **Make sure to upload any code you use to answer the tasks** (Implementation tasks include 2, 4, 5, and 6 for Benders and task 3 for Progressive Hedging).

The solvers Gurobi or HiGHS can be used for Benders Decomposition; however, Gurobi should be used for the progressive hedging tasks.

Benders Decomposition

1. Formulate a mathematical model that can be used to solve the company's problem. Your formulation should clearly define all decision variables, the objective function, and all relevant constraints.
2. Solve your model using a solver of your choice. Report the optimal objective value, identify which suppliers are selected in the optimal solution, and specify the amount of capacity reserved from each selected supplier.
3. Explain why the formulation is suitable for Benders decomposition and derive an appropriate Benders reformulation. Clearly present the master problem, the subproblem(s), and the explicit forms of any optimality or feasibility cuts.
4. Using a language of your choice, implement Benders decomposition to solve your problem, dynamically generating any optimality or feasibility cuts. Report the optimal solution obtained via your Benders implementation and state the number of iterations required to reach convergence.
5. Repeat Question 4, but now assume that all demand **must** be met. In such a setting, infeasible primal subproblem forms (unbounded dual subproblem formulations) are possible. Demonstrate that your Benders implementation can identify and add feasibility cuts via extreme rays.
6. Compare a single-cut version of Benders decomposition with that of a multicut. Discuss any observed differences in convergence speed, number of iterations, and nature of the generated cuts. For each variant, include a plot showing how the lower and upper bounds evolve over iterations.

Progressive Hedging

Consider now a simplified version of the problem in which fixed reservation costs and the restriction on the number of suppliers are removed. In this setting, the company needs only to determine how much capacity to reserve at each supplier. The only costs incurred are the per-unit capacity reservation costs, the scenario-dependent operating costs, and the penalties associated with any unmet demand.

1. Explain the core idea behind the Progressive Hedging (PH) algorithm for two-stage stochastic optimization. Describe how scenario subproblems are constructed and how non-anticipativity of first-stage variables is enforced through the PH updates.
2. Formulate the scenario-specific PH subproblems for the simplified stochastic capacity reservation problem. Clearly show how the first-stage capacity decisions and second-stage allocation decisions appear in the PH framework, and derive the update rule for the consensus first-stage capacity vector.

3. Implement the PH algorithm for the simplified stochastic capacity-reservation problem using a programming language of your choice. Report the number of iterations required, the final solution obtained, and any observations about convergence behaviour.